

TARIQ JAMIL

tariq347146@gmail.com | (+92) 335-9759793



<https://github.com/TariqJamilAfridi>



www.linkedin.com/in/tariq-jamil-afridi

Final-year Bachelor of Engineering (B.E.) student in **Computer and Information Systems Engineering** with strong knowledge of **Artificial Intelligence, Machine Learning, Computer Vision, Python programming, Web Development, and Database Systems**. Experienced in developing AI-powered solutions through research internships and academic projects, including an **Intelligent Dairy Farm Health and Milk Production Monitoring System using Computer Vision**. Passionate about building innovative software solutions and applying AI technologies to solve real-world problems.

SKILLS

- ❖ **Python** (Advanced) – Strong understanding of Python programming, including Object-Oriented Programming (OOP), Data Structures and Algorithms (DSA), file handling, exception handling, and software development principles.
- ❖ **Artificial Intelligence (AI) & Machine Learning (ML)** – Knowledge of the complete machine learning pipeline, including data collection, data annotation, preprocessing, model training, evaluation, testing, and deployment.
- ❖ **Computer Vision** – Experience in image and video processing, object detection, behavior analysis, and computer vision applications using AI/ML techniques.
- ❖ **Front-End Development** – Strong knowledge of HTML, CSS, JavaScript, responsive web design, and user interface development.
- ❖ **Back-End Development** – Experience developing backend applications using JavaScript (Node.js), PHP, and Python.
- ❖ **Databases** – Knowledge of SQL and Oracle Database, including database design, queries, normalization, and data management.

EXPERIENCE

- ❖ **National Center of Artificial Intelligence (NCAI)**
Research Intern | Final Year Design Project (FYDP)
 - Working under the supervision of **Professor Muhammad Khurram** on an AI-powered research project titled "**Intelligent Dairy Farm Health and Milk Production Monitoring System Using Computer Vision**".
 - Developing computer vision and machine learning models to monitor cattle behaviours (i.e., standing, lying, feeding, and drinking) and milk production.
 - Performing data collection, annotation, preprocessing, model training, testing, and performance evaluation.
 - Implementing real-time monitoring and alert systems for livestock management.

ACADEMIC PROJECTS

- ❖ **Intelligent Dairy Farm Health and Milk Production Monitoring System Using Computer Vision (FYDP)**
 - Designed and developed an AI-based cattle health monitoring system using Computer Vision and Machine Learning techniques.
 - Analyzed cattle activities such as standing, lying, drinking, and movement patterns using multiple camera feeds.
 - Applied machine learning and computer vision algorithms to identify abnormal behaviours and health-related issues.
 - Developed a real-time monitoring solution that provides alerts and insights to livestock owners for early disease detection and improved farm management.
- ❖ **Additional Python Projects**
 - Developed multiple academic projects using Python focusing on Object-Oriented Programming (OOP), Data Structures and Algorithms (DSA), and software development concepts.
 - Implemented practical solutions involving data processing, automation, and problem-solving techniques.

CERTIFICATIONS

- ❖ **AI & ML Internship Certificate** (8 Weeks) – National Center of Artificial Intelligence (NCAI), Karachi (Supervisor: **Prof. Muhammad Khurram**).
- ❖ **Responsive Web Design Certification** – FreeCodeCamp.
- ❖ **AI for Beginners Certification** – HP LIFE | HP Foundation.
- ❖ **Winner** (1st Place) – **AI/ML Hackathon**, CIS Community, NED University – Best project in AI/ML category.

NED Scholars Organization | Zoom Operations Lead (2022 – Present)

- ❖ Active **volunteer** in **NED Scholars**, a nonprofit organization registered in the **USA** and **Pakistan**, supporting STEM education, scholarships, and mentorship programs.
- ❖ Led the **Zoom Operations Team**, managing technical arrangements for online and hybrid events, webinars, and workshops.
- ❖ Progressed from team member to team lead, demonstrating **leadership, communication, and event management** skills.